The geometric branches of iterated cubic halving

Human Compression project

This note supplies the geometric arguments behind Section 5 and Figures 5–8 of the inspected preprint of Piedade–Tranchida [1]. The constructions are theirs. We use the previously reviewed component formulation of halving [3], including its precise residual-connectivity condition, and the separately reviewed affine group calculation [5]. In particular a presentation or a Coxeter diagram will not be used to infer residual connectivity. All assertions about the source refer to arXiv:2405.19050v1, not to an uninspected later journal edition.

1 The cubical model and the retained halving theorem

Let $n \geq 3$, $s \geq 2$, and $k \in \{1, 2, n\}$. In the usual cubical tessellation of $\mathbb{R}^n$ put

$$D = \{v \in \mathbb{Z}^n : \textstyle\sum v_i \equiv 0 \pmod 2\}, \qquad L = D \cap \Lambda_k,$$

$$\Lambda_1 = s\mathbb{Z}^n, \quad \Lambda_2 = sD, \quad \Lambda_n = s(2\mathbb{Z}^n + \mathbb{Z}\mathbf{1}).$$

A cubical face is a product of integral singleton coordinates and unit intervals. Its dimension is its number of interval coordinates. The original toroid has objects the Λ_k-orbits of faces of dimensions $0, \ldots, n$, with incidence induced by containment. We retain the classical assertion that this is the regular polytope of Theorem 5.1 of [1], attributed there to [2, Theorem 6D4]. Its usual marking, acting on column coordinates, is

$$a = \tau_{e_1} z_1, \quad s_1, \ldots, s_{n-1}, \quad z_n,$$

where s_i swaps coordinates $i, i+1$ and z_i changes the sign of coordinate i. This is the standard affine realization, not a new classification of cubic toroids.

We recall exactly the halving result used here. A selected vertex–edge truncation is required to be a simple graph Q (B1). For every other object, incidence with an edge must be equivalent to incidence with both endpoints (B2). For a flag F of other objects let Q_F be its common graph shadow. If Q is nonbipartite, the additional condition is

$$\text{all singleton shadows } Q_x \ (x \in F) \text{ nonbipartite} \implies Q_F \text{ nonbipartite.} \tag{PC}$$

In the component construction one takes a component of the bipartite double cover of Q, and lifts every other object by the components of the double cover of its shadow. The resulting incidence system is a geometry; it is residually connected when Q is bipartite, and, when Q is nonbipartite, exactly when PC holds. Thinness and chamber transitivity are preserved. These statements, including the residue isomorphisms used to compute the diagram, are the retained results of [3]. The regular coset identification and full type-preserving automorphism group are supplied by [4]. Neither result permits one to omit PC in a nonbipartite step.

Write H_{ab} for the geometry after a operations at the left end and b at the right end, with H_{00} the original toroid. The notation will only be used for the sequences proved to exist below. Choices of the two equivalent leaves at a fork are identified by a type permutation. A "second operation at the left" means the leaf $(0, 2)$ after the first operation $(0, 1)$.

2 Small translations and local cubical incidence

The following elementary separation facts keep the quotient arguments explicit. Put $R = \{\pm e_i \pm e_j : i \neq j\}$.

Lemma 1. *Every nonzero vector of Λ_k or L has a coordinate of absolute value at least s. No two vertices of a unit cube are therefore identified. Also:*

1. *$2e_i \in \Lambda_k$ or $2e_i \in L$ occurs exactly when $(k, s) = (1, 2)$.*
2. *If $r, r' \in R$ are distinct and $r - r' \in L$, then $(k, s) = (1, 2)$ or $(k, s) = (2, 2)$. Conversely both parameter pairs allow such a coincidence.*
3. *Suppose neither of these two parameter pairs occurs. If two even vertices of one unit cube differ by a root modulo L, they differ by that root literally. Their unique square is a face of the cube. If two even vertices at root distance have a common odd unit-step neighbor modulo L, that neighbor is one of the other two corners of this square, after a common lift.*

The corresponding statements hold in the dual cubical grid of cube centers, translated by $\frac{1}{2}\mathbf{1}$.

Proof. All lattices are contained in $s\mathbb{Z}^n$. For (1), it remains only to check $s = 2$. The vector e_i is neither in D nor in $2\mathbb{Z}^n + \mathbb{Z}\mathbf{1}$ for $n \geq 3$.

For (2), every coordinate of $r - r'$ has absolute value at most two. Thus $s = 2$. If $r - r'$ is even coordinatewise, the supports of r, r' are equal, and its nonzero support has size at most two. In $2D$ a single coordinate ± 2 is excluded; a pair $\pm 2, \pm 2$ is allowed. In $2(2\mathbb{Z}^n + \mathbb{Z}\mathbf{1})$ neither a one-coordinate nor a two-coordinate vector of this form occurs, since $n \geq 3$. Passing to L cannot create a coincidence. Examples are $e_1 + e_2$ and $-e_1 + e_2$ for $(1, 2)$, and $e_1 + e_2$ and $-e_1 - e_2$ for $(2, 2)$.

For (3), lift the cube and its two vertices p, q. In $q - p - r \in L$, every coordinate has absolute value at most two. For $s \geq 3$ it is zero. For $s = 2$, a nonzero even coordinate can occur only on the two coordinates supporting r, and the remaining allowed lattice $2(2\mathbb{Z}^n + \mathbb{Z}\mathbf{1})$ contains no such nonzero vector. Hence $q - p = r$. The square has exactly these two varying coordinates, so is unique and contained in the cube.

For the common-neighbor assertion, first lift the neighbor w and one endpoint p so that $w - p$ is a unit vector. The other incidence gives $q - p \equiv \alpha + \beta \pmod{L}$, where α, β are signed unit vectors. This sum is zero, a root, or $\pm 2e_i$. The zero case cannot equal a root modulo L. Two roots are equal by (2). Finally $\pm 2e_i - r$ cannot belong to $s\mathbb{Z}^n$ for any $s \geq 2$: at least one coordinate belonging to the other coordinate of r has value ± 1. Thus the sum is the same root, and the two possible intermediate vertices are exactly the other square corners. Translation of the entire argument gives the dual statement. □

We will also use the following form of local lifting. Fix a unit cube C, or a cube in the dual grid. Its vertices and subfaces inject into the quotient: a nonzero period cannot move a vertex or a subface of C to another vertex or subface of C, since the coordinate differences have absolute value at most one. An incident chain of face orbits has a common lift by first fixing its largest face and then lifting the smaller faces inside it. These lifts are unique. Incidences between subfaces of that largest face are consequently literal containments. When a vertex and a chain of faces form a flag, the vertex has its unique lift inside the smallest face. These observations apply to primal and dual cubes, and will be used together with Lemma 1 when a flag has two distinct vertex types.

Proposition 2. *The original left and right leaves satisfy B1 and B2 exactly when $(k, s) \neq (1, 2)$. After the first left operation the geometry H_{10} has the following literal model modulo L:*

0	*even integral vertices*
1	*odd integral vertices*
$i \geq 2$	*original i-cubes.*

Incidence between types 0 and 1 is unit-step adjacency; all incidences with cubes and between cubes are containment. This geometry is a regular hypertope.

Proof. Unit-step edges can become parallel in the quotient only if two distinct signed unit vectors differ by a period. Because periods are in $s\mathbb{Z}^n$, this is exactly the case $2e_i \in \Lambda_k$, settled by Lemma 1(1). To prove B2, lift a cube and two of its vertices. If their difference is a unit vector modulo Λ_k, the error has coordinates of absolute value at most two and can have a nonzero even coordinate only on that single axis. The only possible period is again $\pm 2e_i$ at $(k, s) = (1, 2)$. Otherwise the edge is a literal edge of the cube. The forward implication is containment. Self-duality proves the right-end assertion.

The parity of the coordinate sum colors the infinite grid. Its color-preserving period subgroup is exactly L. The component construction on the quotient by Λ_k is consequently the displayed colored-vertex construction modulo L. More explicitly, if $\Lambda_k \subset D$ the quotient grid is bipartite and no objects are doubled; if an odd period exists, there are two cosets of L in Λ_k, giving the two vertex lifts and the two compatible lifts of each cube. The three displayed lattices lie in D exactly when ks is even; an odd period otherwise supplies an odd closed walk in the quotient grid. Every cube shadow is an ordinary cube graph and is bipartite; higher flags are chains of cubes and their shadows are the smallest cubes. Thus PC holds, including in the globally nonbipartite case. The retained halving theorem gives thinness, residual connectivity and regularity. □

3 A second operation at one end

Proposition 3. *In H_{10} the leaves $(0, 2)$ and $(1, 2)$ satisfy B1 and B2 exactly when $(k, s) \neq (2, 2)$, assuming $(k, s) \neq (1, 2)$ already. Under this condition the second-left graph is nonbipartite, PC holds, and its halving H_{20} is a regular hypertope. Its objects modulo L are*

$0, 2$	*two separate copies of the even integral vertices*
1	*odd integral vertices*
$i \geq 3$	*original i-cubes.*

The two even-vertex types are incident when their underlying vertices differ by a root in R; even and odd types are incident at unit distance; incidences with cubes and between cubes are containment.

Proof. Each square has exactly two even corners. Its resulting edge has a root as difference vector. Conversely a root specifies a unique square. Lemma 1(2) therefore proves simplicity except at the stated small period. For B2 at a higher cube use part (3) of that lemma. At an odd vertex w, the even neighbors are the $2n$ points $w \pm e_i$. They are distinct by part (1). Two such neighbors give a square through w exactly when they are on different axes. Two opposite neighbors on the same axis cannot differ by a root modulo L, by the last argument in part (3). This proves B2.

At $(k, s) = (2, 2)$ the squares on coordinates 1,2 with respective corner sets $\{0, e_1, e_2, e_1 + e_2\}$ and $\{0, -e_1, -e_2, -e_1 - e_2\}$ are distinct, but have the same two even corners modulo $2D$. Hence B1 fails. B2 fails too: the odd vertex e_1 is adjacent to both these even corners but is not a corner of the second square modulo $2D$.

The root graph is connected because R spans D, and contains the triangle $0, e_1 + e_2, e_1 + e_3$. Its three vertices remain distinct. The graph shadow of an m-cube, $m \geq 3$, is its demicube: one parity class of its corners, adjacent at Hamming distance two. It is connected and contains a triangle. Indeed, any even difference can be removed two coordinates at a time, and the three corners with difference supports $\varnothing, \{1, 2\}, \{1, 3\}$ form a triangle (with a translated choice for the other parity). The shadow at an odd vertex is the cross-polytope graph on $\{w \pm e_i\}$, with edges between nonopposite pairs. It is connected and contains a triangle for $n \geq 3$.

A higher flag is either a chain of cubes, whose common shadow is the demicube of its smallest member, or such a chain together with an odd vertex w. In the latter case its even neighbors inside the smallest m-cube are the m corners obtained from w by a single allowed step; they form K_m, with $m \geq 3$. If there is no cube the shadow is the full cross-polytope graph. Local lifting justifies these descriptions in the quotient. All of them are nonbipartite, proving PC. Thus every higher object has just one lift in the new component construction, while the even vertices acquire two copies. This gives precisely the asserted model and proves its regular-hypertope conclusion. □

There is a useful consequence of this proof. Any flag of just the three lower types in H_{20} has a common lift whose vertices span an original face of dimension at most two. A single vertex spans a point; an even–odd pair spans an edge; a pair of even types spans their unique square. If the third, odd, type occurs, Lemma 1 puts it at one of the other corners of that square. The span is unique. If a cube is also prescribed, the span lies inside its unique lift. For H_{10} the corresponding lower flags span at most an edge. We call this the *small-face property* below.

4 Operations at the opposite end

Cube centers form the dual grid $\frac{1}{2}\mathbf{1} + \mathbb{Z}^n$. Color a center $c = \frac{1}{2}\mathbf{1} + v$ by the parity of $\sum v_i$. This is well-defined modulo L. For a primal d-face F, the centers of the n-cubes containing F form the vertices of a dual cube of dimension $n - d$. A facet between two such cubes is a dual edge. These are literal local statements, also after quotienting by L, by the local-lifting observation.

Proposition 4. *One may perform the first right operation after H_{10} for all $n \geq 3$, and after H_{20} for $n \geq 4$. The selected graph is bipartite, B1 and B2 hold, and the resulting geometries H_{11} and H_{21} are regular hypertopes. Their two upper types $n, n-1$ are respectively the two colors of original n-cube centers. These types are incident precisely when the cubes share a facet. Lower objects retain their earlier model, and are incident with an upper cube when they are contained in it. The cubes previously of dimension $n-1$ are removed.*

Proof. In both cases the original n-cubes and $(n-1)$-cubes are still present: for H_{20} this uses $n - 1 \geq 3$. Their graph is the dual unit-step grid modulo L. It is simple by Lemma 1(1), and bipartite by the displayed coloring. B2 for a lower original cube is the primal/dual version of the edge argument in Proposition 2. For a lower vertex, including either copy of an even vertex, it is the same argument for the dual n-cube around that vertex. No other object types occur. Every higher shadow is a connected bipartite cube graph: for a flag without an original cube, use its small-face span; when there is a chain of original cubes, use the largest one. Their codimension is at least two. The component construction therefore splits the center graph into its two colors, does not duplicate any other object, and has precisely the model asserted. Residual connectivity follows from the bipartite case of the retained theorem, without an additional PC assumption. □

Proposition 5. *A second right operation is possible in H_{11} when $n \geq 4$ and in H_{21} when $n \geq 5$, provided $(k, s) \neq (2, 2)$. Its graph is nonbipartite, B1 and B2 hold, and PC holds. It therefore gives H_{12} (dual to H_{21}) and H_{22}, respectively, as regular hypertopes. The two new upper types $n, n-2$ are separate copies of the even centers, incident at root distance; type $n-1$ consists of odd centers. All remaining objects keep their earlier meanings.*

Proof. The objects of type $n - 2$ are still original $(n-2)$-faces under the rank assumptions. They have four incident n-cubes, two with even centers. Thus the selected graph is the root graph on the even dual vertices. The proofs of simplicity and B2 are exactly Lemma 1 in the dual grid, applied separately to each actual higher object: a remaining original face has a dual cube as its shadow, a lower vertex has a dual n-cube, and an odd center has its cross-polytope neighbor graph. Different copies of a lower vertex have the same singleton shadow, which causes no problem for B2.

For completeness consider an arbitrary higher flag, not just its singletons. First suppose it contains no odd center. If it contains original faces, their largest member F determines the common dual cube, since all the lower vertices are in F. Its dimension is $n - \dim F \geq 3$, because type $n-2$ is being halved and the remaining face dimensions are at most $n-3$. If the flag contains only lower vertex types, their small-face span has dimension $d \leq 1$ in H_{11} and $d \leq 2$ in H_{21}. The common even-center shadow is the demicube of a dual cube of dimension $n - d \geq 3$. This also covers a singleton. An empty flag has the whole root graph, which is connected and nonbipartite.

Now suppose an odd center c is in the flag. Lift its primal n-cube. Every prescribed lower vertex and every prescribed primal face lifts inside this cube. The small-face property and local lifting make these lifts mutually compatible. Let F be their largest face, or their small-face span if

there is no prescribed face. A neighboring even-center cube contains F exactly when its common facet with the cube at c contains F. There are $n - \dim F$ such facets, so the shadow is the clique $K_{n-\dim F}$. Its size is at least three by the same bounds. If there are no other objects, the shadow is the full cross-polytope graph. All these graphs are connected and nonbipartite. This proves PC for every flag. The component construction now gives exactly the stated duplicate even-center model. The retained theorem proves the conclusion. □

These arguments also identify the order of the operations. Where both are defined, doing the first right operation before the second left operation yields the same H_{21} incidence system. One can either apply Proposition 5 by duality to H_{11}, or compare the displayed models: both have two even-vertex copies, odd vertices, the remaining primal cubes, and the two center colors, with exactly the incidences already specified. The same comparison identifies the two paths to H_{22}. Thus an arrow in a diagram of constructions does not rely merely on equality of the group orders.

5 Exact diagrams and exhaustion of leaf operations

The regular-marking theorem in [4] replaces a selected leaf generator r by rqr, where q is its neighbor. Therefore the markings of the models just proved have the following uniform form:

$$(\ell_a, s_1, \dots, s_{n-1}, u_b),$$

where

j	ℓ_j	u_j
0	$\tau_{e_1} z_1$	z_n
1	$\tau_{e_1+e_2} z_1 z_2 (1\,2)$	$z_{n-1} z_n (n-1\,n)$
2	$\tau_{e_1+e_3} z_1 z_3 (1\,3)$	$z_{n-2} z_n (n-2\,n)$.

Here $(i\,j)$ denotes a coordinate swap. The second row is the first conjugation at each end; conjugating the next inward swap gives the third. This also checks all overlap cases in small rank. Since the geometries are thin and residually connected, these product orders are the actual rank-two residue gonalities.

We give indexed edge tables to make the small-rank diagrams unambiguous. Unlisted pairs have order two. Let

$$P = \{\{i, i+1\} : 1 \le i \le n-2\}.$$

Its edges all have label three. In the next table, an edge listed in the last column has label four; all other listed edges have label three. Repetitions in a union do not add another edge.

state	edges of label three	edges of label four
00	P	$01,\ (n-1)n$
10	$P \cup \{02\}$	$(n-1)n$
$11,\ n=3$	$P \cup \{02, 03, 13\}$	$\varnothing$
$11,\ n \ge 4$	$P \cup \{02, (n-2)n\}$	$\varnothing$
$20,\ n=3$	$P \cup \{01, 02\}$	$03, 23$
$20,\ n \ge 4$	$P \cup \{01, 02, 03\}$	$(n-1)n$
$21,\ n=4$	$P \cup \{01, 02, 03, 04, 24\}$	$\varnothing$
$21,\ n \ge 5$	$P \cup \{01, 02, 03, (n-2)n\}$	$\varnothing$
$22,\ n=5$	$P \cup \{01, 02, 03, 05, 25, 35, 45\}$	$\varnothing$
$22,\ n \ge 6$	$P \cup \{01, 02, 03, (n-3)n, (n-2)n, (n-1)n\}$	$\varnothing$

For example, 02 means the unordered pair $\{0, 2\}$. The states 21 and 22 only occur at the ranks in the table.

Here is a direct verification of the product orders, independent of any drawn diagram. Interior swaps on disjoint supports commute; consecutive ones have product order three. A signed swap on coordinates i, j has product order three with a swap sharing exactly one coordinate, and commutes

with a disjoint swap. The two opposite signed swaps on the same pair commute. A sign change on one coordinate of a signed swap has product order four. The same statements apply to the affine left reflections after translating their common fixed intersection to the origin. In the few overlapping left–right pairs one can check the displayed maps directly: for $n = 3$, $\ell_1 u_1$ has order three and $\ell_2 z_3$ has order four; for $n = 4$, ℓ_2 and u_1 have order three; for $n = 5$, ℓ_2 and u_2 have order three. The linear parts already have these orders. Thus the orders cannot decrease after a translation quotient. Pairs declared commuting are commuting as affine maps, so the table holds for every allowed period.

Theorem 6. *Exclude $(k, s) = (1, 2)$. Up to duality and exchange of the two equivalent fork leaves, the complete sequence of admissible leaf halvings has states*

$$00 \longrightarrow 10 \longrightarrow 11,$$

and, if also $(k, s) \neq (2, 2)$,

$$10 \longrightarrow 20, \qquad 20 \longrightarrow 21 \longleftarrow 11 \quad (n \geq 4), \qquad 21 \longrightarrow 22 \quad (n \geq 5).$$

The initial arrow uses the bipartition construction if ks is even and the partition construction if ks is odd. The arrows at the first opposite end are bipartition operations. Each second operation at an end is a nonbipartite partition operation. For $n = 3$, states 20 and 11 are terminal; for $n = 4$, state 21 is terminal; for $n \geq 5$, state 22 is terminal. Every permitted state is a regular hypertope, with the exact diagram in the table.

Proof. Existence, nondegeneracy, parity and PC for all arrows have been proved in Propositions 2–5 and their dual forms. The original left endpoint reflection a exchanges vertex parity and the two left fork markings. At the other end z_n preserves vertex parity and exchanges center parity; it conjugates the two right fork markings. Whenever a second operation at the opposite end is already present, the stated rank bounds make these endpoint reflections commute with its altered marker. Thus these equivalences still apply at that stage. The explicit models identify both orders of independent operations.

It remains to justify exhaustion. The table shows that 00 has only its two endpoints as leaves. State 10 has its two left fork leaves and its right endpoint. At $(k, s) = (2, 2)$ the former fail B1 by Proposition 3; the latter still works. State 20 has no leaf when $n = 3$, and otherwise only the right endpoint. State 11 has no leaf when $n = 3$; otherwise its four fork leaves lead, up to the symmetries already described, to 21. State 21 has no leaf when $n = 4$, and otherwise just the two right fork leaves. Finally every vertex in state 22 has degree at least two. These are exactly the states and arrows stated. At the excluded second-step period, the same dual root coincidence prevents the fork operations in 11 as well. There is therefore no additional leaf operation omitted from the classification. □

This is the precise interpretation of the families in Figures 5–8: the second-step arrows require the extra small-period exclusion; the last two graphs are the overlapping-end cases in ranks four and five, not disjoint copies of the large-rank picture. The table specifies every bond even where a reproduced figure is difficult to read. The proof supplies the nondegeneracy and PC checks which the final paragraph of Section 5.3 leaves implicit. It does not restore the insufficient exact presentations in Propositions 5.8–5.13: the corrected marked groups and all missing period relations are treated separately in [5].

6 Scope and attribution

The toroids, the halving operation, the proposed diagram families, and their attribution to earlier polytope and diagram geometry are retained from the source. The new argument here is an explicit coordinate verification of every allowed geometric branch, using the previously reviewed parity-shadow criterion rather than the source's overbroad residual-connectivity claim. It is a Section 5 recovery component. It makes no whole-original admission, numerical compression claim, new conjecture claim or formalization claim.

References

[1] C. A. Piedade and P. Tranchida, *Constructing new geometries: a generalized approach to halving for hypertopes*, arXiv:2405.19050v1, 29 May 2024, Section 5.

[2] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Encyclopedia of Mathematics and its Applications 92, Cambridge University Press, 2002, Theorem 6D4 (retained through the source's Theorem 5.1).

[3] Human Compression project, *Geometric halving through graph covers and parity-compatible shadows*, reviewed phase27 parity theory, Sections 1–7.

[4] Human Compression project, *Stabilizers in geometric halving*, reviewed N014 coset-identification packet. Exact dependency hashes accompany this note.

[5] Human Compression project, *Affine groups after one and two halvings*, reviewed N014 affine-halving packet. Exact dependency hashes accompany this note.